

CLINICAL DEEP DIVE · COMPANION REPORT

Dazodalibep in Sjögren’s Disease

A Plain-Language Guide to the Phase 3 Forecast — explaining the Phase 3 efficacy forecast for a general reader, slide by slide

PRIMARY FORECAST	NCT06104124 — systemic disease activity trial
SUPPORTING ASSESSMENT	NCT06245408 — symptom burden trial
BASED ON PRESENTATION	“Dazodalibep in Sjögren's disease: forecasting the Phase 3 readout, and what could move it” (26 slides)
ANALYSIS CUTOFF	September 15, 2026
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This report explains, in plain language, every slide of the source presentation — the metrics, the charts, and the reasoning — and closes with an explicit conclusion on whether the forecast trial outcome is positive or negative, and why. No figures have been altered from the source analysis; only explanatory language has been added.

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READING GUIDE

How to Read This Report

This report is a plain-language companion to a 26-slide equity-research presentation forecasting the outcome of Amgen's Phase 3 clinical trial program for **dazodalibep**, an experimental drug for Sjögren's disease. It is written so that a reader with no biotech or statistics background can follow the evidence and the reasoning, while still preserving every metric, caveat, and source citation from the original analysis.

The report is organized into four parts:

- **Background** — a short primer on the disease, the drug, the two trials, and the key terms used throughout (read this first if you are unfamiliar with the subject matter).
- **Part I** — a slide-by-slide walkthrough of the main 20-slide presentation, in the original order, with each slide's data, charts, and reasoning explained.
- **Part II** — a walkthrough of the presentation's six backup/appendix slides, which contain supporting detail (registry data, baseline tables, statistical methodology, and sources).
- **Conclusion** — a direct answer to the question “is the forecast positive or negative,” with the full reasoning laid out, followed by report-level appendices (full glossary, complete source list, and a note on this report's own methodology and limitations).

Every metric in this report is carried over unchanged from the original presentation — nothing has been recalculated, estimated, or rounded differently. Where the original analysis flags a number as an assumption, an estimate, or unconfirmed, this report preserves that flag rather than smoothing it over.

SUMMARY

Executive Summary

For readers who want the bottom line before the detail: this section restates the presentation's own headline conclusion (originally Slide 2), in the same plain language used throughout this report.

The analysis expects the **systemic trial (NCT06104124)** to succeed on its primary endpoint — a statistically significant improvement in ESSDAI, the clinician-scored measure of disease activity — but explicitly flags meaningful residual risk that this could still fail. For the **symptomatic trial (NCT06245408)**, the call is softer: a positive lean on the older, tested symptom measure (ESSPRI), tempered by real uncertainty around a new, never-before-used co-primary measure (DASPRI) and an undisclosed rule for what counts as overall success.

This conclusion rests on three supporting pillars — a strong Phase 2 result, biological evidence the drug is active, and a competitor drug that has already shown the same type of endpoint can be won in Phase 3 — balanced against several disclosed risks: a bigger and longer trial that could dilute the effect, unknown statistical testing rules, real sensitivity to data noise, an untested second symptom instrument, and a small but real safety signal from Phase 2. All of this evidence is walked through slide by slide in Part I and Part II below, and synthesized into a final call in the Conclusion.

PRIMER

Background & Key Terms

The disease: Sjögren's disease

Sjögren's disease (also called Sjögren's syndrome) is a chronic autoimmune disease in which the immune system mistakenly attacks the body's own moisture-producing glands — primarily the glands that make tears and saliva. This produces the disease's hallmark symptoms: persistent dry eyes and dry mouth. Beyond dryness, many patients also experience significant fatigue and joint pain, and in more severe cases the disease can become “**systemic**” — meaning it extends beyond the glands to affect joints, skin, blood counts, kidneys, lungs, and other organs. There is currently no cure; existing treatments mostly manage symptoms (for example, artificial tears and saliva substitutes) rather than address the underlying overactive immune process. This is the unmet medical need dazodalibep is trying to address.

The drug: dazodalibep

Dazodalibep is an experimental biologic (an antibody-like protein, not a small-molecule pill) designed to calm a specific piece of immune-system overactivity implicated in Sjögren's disease. The full mechanism is explained in detail at Slide 5, but in short: it blocks a “costimulatory” signal called CD40L, which helps drive the immune system to attack the body's own tissue. By interrupting that signal, the drug aims to quiet the underlying disease process rather than just treating individual symptoms.

The two trials this report is about

Amgen is running two separate, simultaneous Phase 3 trials, each targeting a different kind of Sjögren's patient:

- **NCT06104124 (“the systemic trial”)**: enrolls 651 patients with moderate-to-severe systemic (organ-level) disease activity, and measures whether the drug improves a clinician-scored activity index (ESSDAI) after 48 weeks. This is the trial the report treats as the primary forecast.
- **NCT06245408 (“the symptomatic trial”)**: enrolls 434 patients with high day-to-day symptom burden but limited organ-level activity, and measures whether the drug improves two patient-reported symptom scores (ESSPRI and DASPRI) after 48 weeks. This trial gets a lighter, more qualified treatment in the report.

Both trials build on an earlier Phase 2 trial, published in 2024, which forms the evidentiary foundation for most of the forecast in this report.

The key measures, in plain terms

Term	What it is
ESSDAI	A disease-activity score filled out by a <i>clinician</i> during an exam. It grades how active the autoimmune disease currently is across body systems (joints, glands, blood, skin, kidneys, lungs). Higher score = more active disease; a bigger drop = more improvement.
ESSPRI	A symptom score filled out by the <i>patient</i> , covering dryness, fatigue, and pain. Higher score = worse symptoms; a bigger drop = more improvement.
DASPRI	A newer patient-reported symptom diary, developed to align with U.S. FDA guidance on patient-reported outcome measures. It has never been used in any dazodalibep trial before — this is one of the report's central identified risks (see Slide 18).

These measures are related but not interchangeable — a patient can score poorly on one and well on the other, because they capture different things (organ-level clinical activity versus how the patient actually feels day to day). This distinction, covered at Slide 4, is central to why this report treats the two trials as separate bets rather than a single combined outcome.

PART I

Main Presentation Walkthrough

A slide-by-slide explanation of the 20-slide core presentation — the disease and trial background, the Phase 2 evidence, the Phase 3 design changes, the statistical stress-tests, and the safety review — in the original presentation order.

PART I · SLIDE 1 OF 26

1 Title & Framing

This is the cover slide of the presentation. It states the subject — forecasting the outcome of Amgen's Phase 3 program for **dazodalibep**, an experimental drug for Sjögren's disease — and sets the scope for everything that follows.

What it establishes:

- The **primary forecast** is for trial **NCT06104124**, which measures systemic (whole-body) disease activity. This is the trial the report treats as the main event.
- A **supporting assessment** covers trial **NCT06245408**, which measures patient-reported symptom burden (dryness, fatigue, pain). This trial gets a lighter, more qualified treatment throughout.
- The analysis is dated to a cutoff of **September 15, 2026** — i.e., it uses only information that was publicly available up to that date.

No results are previewed here. The slide is deliberately neutral: it names the two trials and the analysis date, and nothing about direction of outcome.

PART I · SLIDE 2 OF 26

2 Executive Conclusion

This is the headline slide of the whole presentation — the author's bottom-line call, stated up front in equity-research style (conclusion first, evidence after).

The call: the analyst expects **NCT06104124** (the systemic trial) to meet its primary endpoint — i.e., to show a statistically significant improvement in **ESSDAI**, the clinician-scored measure of disease activity — but flags **meaningful residual risk** that this could still go wrong.

Three pieces of evidence are offered as the basis for optimism:

1. **Direct clinical evidence:** in the earlier (Phase 2) trial, patients on dazodalibep improved on ESSDAI by 2.2 points more than patients on placebo, a difference that was statistically significant ($p=0.0167$).
2. **Biological coherence:** the drug lowered two immune markers in the blood (CXCL13 and rheumatoid factor), which is consistent with the drug actually doing what it is designed to do at a biological level. A related drug aimed at the same pathway (iscalimab) shows similar evidence.
3. **Confirmatory precedent:** a competitor drug in the same disease (ianalumab) has already shown, in its own Phase 3 trials, that an ESSDAI benefit *can* be detected at the week-48 timepoint — meaning the endpoint itself is provable in a Phase 3 setting, not just a Phase 2 artifact.

For the supporting symptomatic trial, the call is softer: a “**positive ESSPRI lean**” — meaning the symptom evidence points positive — but conviction is capped by uncertainty around a second, untested endpoint (DASPRI) and an undisclosed rule for what counts as overall trial success (explained fully at Slide 18 below).

The main risk, stated plainly: if the true effect size shrinks in the larger, longer Phase 3 trial, or if the data turns out noisier than expected, or if the statistical bar is set stricter than in Phase 2, an encouraging signal could tip into a technically negative result. This isn't a hedge for its own sake — the middle third of this report (Slides 15–17) is built specifically to quantify how easily that could happen.

Source: analyst assessment based on the project's underlying analysis files. The report is explicit that no numerically calibrated probability (e.g., “70% chance of success”) is being claimed — the forecast is qualitative and directional, and is stress-tested later in the deck.

3 Two Distinct Clinical Questions

Before getting into evidence, this slide draws a line that matters for everything that follows: the program is not one trial asking one question. It is **two separate Phase 3 trials**, enrolling different patients and testing different things at the same week-48 timepoint.

	Systemic — NCT06104124	Symptomatic — NCT06245408
Who is enrolled	Patients with moderate-to-severe systemic disease activity	Patients with high symptom burden but limited systemic (organ-level) activity
Entry requirement	ESSDAI ≥ 5	ESSPRI ≥ 5 and ESSDAI < 5
Enrollment size	651 patients	434 patients
What's being measured (primary outcome)	Change in ESSDAI from baseline	Change in <i>both</i> ESSPRI <i>and</i> DASPRI from baseline
When it's measured	Week 48	Week 48
Status as of the cutoff date	Completed — primary completion July 23, 2026	Active, not recruiting — primary completion estimated October 22, 2026
Treatment in this report	Full efficacy forecast (the main focus)	Directional supporting assessment only

Why this distinction matters for a layperson: think of it as two different clinical trials for two different kinds of Sjögren’s patients — one for people whose disease is attacking their organs and joints (“systemic”), and one for people whose main problem is dryness, fatigue and pain without much organ involvement (“symptomatic”). A drug could plausibly succeed in one and not the other.

An important definitional point the slide makes explicitly: “positive” in this report means the trial hit its statistical bar under the pre-specified primary testing procedure — nothing more. Whether the result is also clinically meaningful, or whether the drug’s benefits outweigh its risks, are separate questions the report addresses later (see Slide 19 on safety and the Conclusion).

Source: ClinicalTrials.gov registry entries for both trials. Full eligibility criteria and testing procedures are governed by the trial protocols, which are not publicly available — the registry is a summary, not the full rulebook.

PART I • SLIDE 4 OF 26

4 Two Related but Separate Tests

This slide explains, in plain terms, *why* a drug could help with one measure and not the other — because ESSDAI and ESSPRI/DASPRI are measuring genuinely different things, scored by different people.

- **ESSDAI** is scored by a *clinician*. It grades disease activity across weighted organ “domains” — joints, glands, blood counts, skin, kidneys, lungs, and so on. It is an objective, exam-based measure of how active the autoimmune disease currently is in the body.
- **ESSPRI** and **DASPRI** are *patient-reported*. They come from a questionnaire or diary the patient fills out about how they feel — specifically dryness (eyes and mouth), fatigue, and pain.

The slide is careful to say these overlap only **partially**, not completely. A patient can feel very dry, fatigued and sore (high ESSPRI) while having relatively quiet organ-level disease activity (low ESSDAI) — and vice versa. Symptoms can arise from several different underlying processes, so calming the immune system enough to move the clinician-scored number does not guarantee it will move the patient-reported number by the same amount, or at all.

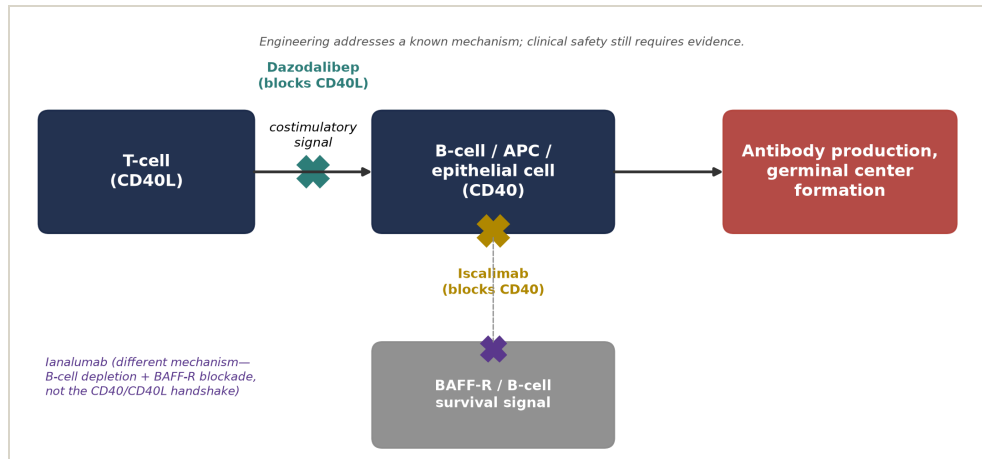
Why this matters for the forecast: it is the biological and statistical reason the report treats the two Phase 3 trials as separate bets rather than one combined outcome. A positive systemic (ESSDAI) result does not automatically imply a positive symptomatic (ESSPRI/DASPRI) result.

Source: Phase 2 baseline characteristics data. The report notes that the specific explanation involving gland function is the analyst's clinical interpretation, not a proven mechanism.

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5 Mechanism of Action: Blocking CD40L

This slide explains, in plain language, *how the drug is supposed to work* at a biological level.



In Sjögren's disease, the immune system is overactive and attacks the body's own glands and tissues. Part of what drives this is a "handshake" between two types of immune cells: a protein called **CD40L** on the surface of T-cells locks onto a receptor called **CD40** on B-cells (and other immune/epithelial cells). This handshake is a key costimulatory signal that helps activate the immune response — including antibody production and the formation of the immune structures that drive autoimmune attack.

Dazodalibep blocks CD40L — it grabs onto the T-cell side of the handshake so the two proteins can never connect. A related drug, **iscalimab**, blocks the same handshake from the other side, by targeting CD40 on the receptor side. Both interrupt the same conversation between immune cells, just from opposite ends. A third competitor drug, **ianalumab**, works completely differently — it depletes B-cells directly and blocks a separate growth signal (BAFF-R), rather than interrupting the CD40L/CD40 handshake at all. This distinction matters later in the report (Slide 13), because ianalumab's Phase 3 results are used as a precedent for whether the same disease-activity endpoint can be won at all — not as a mechanistic twin of dazodalibep.

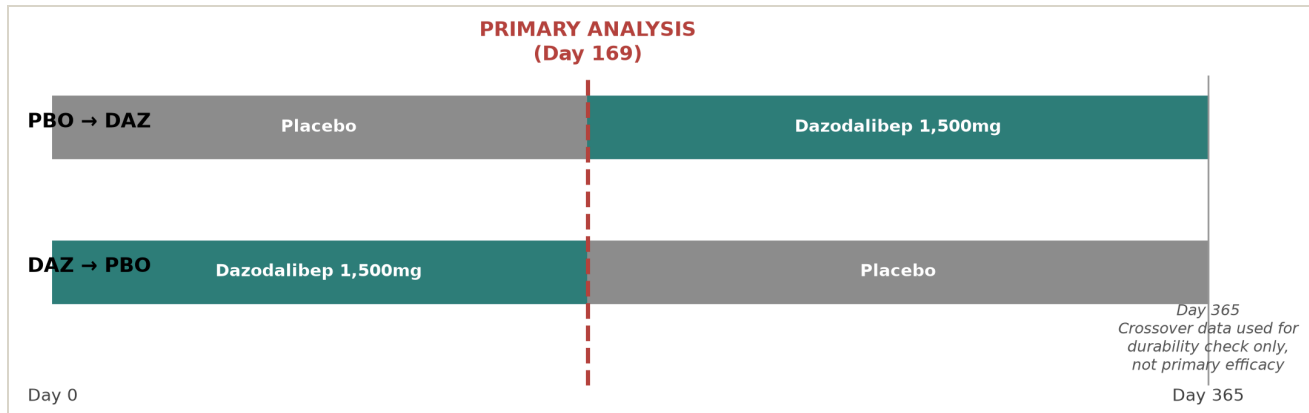
A safety-relevant engineering detail: dazodalibep is built on an "Fc-free" antibody-like scaffold (a Tn3 protein scaffold, not a full antibody). This is a deliberate design choice to avoid a mechanism that caused blood-clotting problems in earlier, first-generation anti-CD40L drugs, where the antibody's Fc portion could cross-link platelets and trigger clots. This detail becomes directly relevant again in the safety discussion at Slide 19.

Source: project background research and the Phase 2 published results (the scaffold is a Tn3 design, not "PASylated" as sometimes mischaracterized elsewhere).

PART I · SLIDE 6 OF 26

6 Phase 2 Trial Design

Before showing any results, this slide explains the design of the earlier (Phase 2) trial that everything else in the middle of the report is built on.



The Phase 2 trial actually contained **two separate randomized populations**, run in parallel — mirroring the systemic/symptomatic split now carried into Phase 3:

- **Population 1 (systemic, ESSDAI ≥ 5 at entry):** 74 patients total, randomized to dazodalibep (36) or placebo (38).
- **Population 2 (symptomatic, ESSPRI ≥ 5 , ESSDAI < 5 , with a minimum stimulated saliva flow requirement):** 109 patients total, randomized to dazodalibep (54) or placebo (55).

Both populations were followed to a **primary readout at day 169** (roughly 5½ months) — ESSDAI change for Population 1, ESSPRI change for Population 2. After that primary readout, the trial **blindly switched** every patient's treatment (placebo patients moved to drug, drug patients moved to placebo) and followed them further to day 365. The report is careful to note that this crossover period is used only as a *durability check*, not as a second independent efficacy test — more on why that distinction matters at Slide 10.

The dosing regimen was a single fixed schedule: **1,500 mg administered intravenously every 2 weeks for 3 doses, then every 4 weeks for 4 doses**. There was no dose-ranging in this study — everyone on active drug got the same dose. (Phase 3, by contrast, tests two different doses — see Slide 11.) Most patients completed the initial stage of the trial (102 of 109 in Population 1; 71 of 74 in Population 2).

Source: the published Phase 2 trial paper and results tables.

7 Phase 2 Result: The Systemic (ESSDAI) Signal

This is the single most important data point behind the “expect positive” call on Slide 2: the actual Phase 2 result for the systemic population.

	Dazodalibep	Placebo
Adjusted mean ESSDAI change at day 169 (± standard error)	−6.3 ± 0.6	−4.1 ± 0.6

How to read this: ESSDAI is a disease-activity score where *higher is worse*, so a larger negative change means more improvement. Both groups got better on average — even the placebo group dropped 4.1 points, which is a well-known pattern in this kind of trial (patients tend to improve somewhat regardless of treatment, partly from closer monitoring and partly from natural fluctuation). What matters is the *difference* between the two groups.

Adjusted difference: −2.2 points in favor of dazodalibep. The reported 90% confidence interval is approximately −3.6 to −0.7, and the p-value is **0.0167**.

A statistical detail worth flagging for a general reader: this trial used a 90% confidence interval (matching a two-sided significance threshold of $\alpha=0.10$) as its pre-specified standard, reflecting the small sample size common in early-phase trials — not the tighter 95% interval used in some other places in this report (for example, the ianalumab comparison on Slide 13). This is called out explicitly wherever it applies, because mixing up a 90% and a 95% interval without noting it would be misleading. For what it’s worth, this particular result is strong enough that it would also have cleared the stricter, more conventional 0.05 significance threshold.

The report notes it is not reproducing the full week-by-week ESSDAI trajectory or exact patient-at-risk counts from the original paper’s figures — only the day-169 headline numbers above.

Source: Phase 2 published results.

8 Responder Analysis: A More Cautious Read

Averages can hide a lot. This slide asks a different, more clinically intuitive question: *how many individual patients* improved by a meaningful amount, rather than just what the group average looked like?

Improvement threshold	Dazodalibep	Placebo	p-value
≥ 3 ESSDAI points	26/36 (72.2%)	22/37 (59.5%)	0.3283
≥ 4 ESSDAI points	24/36 (66.7%)	18/37 (48.6%)	0.1823
≥ 5 ESSDAI points POST HOC	61.1%	35.1%	0.0449

The honest reading here: at every threshold, more dazodalibep patients hit the improvement bar than placebo patients — the direction is consistently favorable. But none of the *pre-specified* thresholds (≥3 or ≥4 points) reaches statistical significance on its own. Only the ≥5-point threshold does, and that comparison was done *after the fact* (“post hoc”), which means it should be treated as suggestive, not confirmatory — it wasn’t one of the tests the trial was designed in advance to answer.

A subgroup exploration is also noted: patients who also had rheumatoid arthritis or lupus (RA/SLE) alongside Sjögren’s showed a larger exploratory effect (adjusted difference roughly –3.3 points) than patients without those overlapping conditions (roughly –1.6 points) — but these are small subgroups within an already-small trial, so this should be read as a hypothesis to watch, not a proven interaction.

On-slide takeaway (verbatim intent): directionally favorable results across all thresholds do not, on their own, establish a robust responder effect or a confirmed subgroup interaction. This slide is the report’s built-in caution against over-reading the headline day-169 average from Slide 7.

Source: Phase 2 published results.

9 Phase 2 Result: The Symptomatic (ESSPRI) Signal

This slide presents the equivalent Phase 2 result for Population 2 — the symptomatic cohort — and it is, numerically, the strongest result anywhere in the Phase 2 program.

Outcome	Dazodalibep change	Placebo change	p-value
ESSPRI total (primary)	−1.80	−0.53	0.0002
Dryness	−1.9	−0.8	0.0066
Fatigue	−1.7	−0.3	0.0022
Pain	−1.8	−0.4	0.0010

How to read this: ESSPRI, like ESSDAI, is scored so that a larger negative change means more improvement. Every single row here favors dazodalibep, and the primary total-score result (p=0.0002) is far stronger, statistically, than the systemic ESSDAI result from Slide 7. The fact that all three individual domains — dryness, fatigue, and pain — moved in the same favorable direction adds consistency to the picture, though the report notes these domain-level breakdowns were not statistically adjusted for the fact that multiple comparisons were being made, so they should be read as supportive consistency checks rather than independently confirmatory results in their own right.

Adjusted primary contrast: −1.27 points, 90% confidence interval −1.82 to −0.73 (again a 90% interval, consistent with Slide 7’s convention).

Why the report still doesn’t call this a slam dunk for Phase 3: the Phase 3 symptomatic trial (NCT06245408) tests a much later timepoint — week 48, not day 169 — and adds a second co-primary endpoint, DASPRI, that has no track record at all in this drug’s development to date. That risk is unpacked fully at Slide 18.

Source: Phase 2 published results.

10 Biomarkers and the Crossover Period

This slide adds two more supporting pieces of evidence: blood-based proof the drug is biologically active, and a look at what happened when patients switched treatments partway through the Phase 2 trial.

Biomarker	Pop 1 ratio (DAZ / Placebo)	Pop 2 ratio (DAZ / Placebo)
CXCL13 (immune activation marker)	0.67 / 1.00	0.56 / 1.06
Rheumatoid factor	0.66 / 1.13	0.54 / 0.94

These are ratios *relative to each arm’s own starting level* (not a direct drug-vs-placebo ratio) — a ratio below 1.0 means the marker went down. On both markers, in both populations, dazodalibep pushed levels down while placebo stayed roughly flat or rose slightly. This is meaningful because CXCL13 and rheumatoid factor are markers of immune overactivity, and this is direct laboratory evidence the drug is doing something biologically, independent of what patients or clinicians reported.

The crossover period (day 169 to day 365): when placebo patients switched onto the real drug, they kept improving (systemic ESSDAI went from -4.1 to -6.3). When drug patients switched onto placebo, they partially backslid (ESSDAI worsened from -6.3 back to -4.4). The symptomatic group’s ESSPRI stayed essentially stable after stopping drug (-1.8 to -1.9). Both patterns — new benefit appearing after starting, and partial loss after stopping — are consistent with a real, active drug effect rather than coincidence.

An important limitation the slide is careful to state: this crossover happened after the trial was unblinded to the primary result, so it is not equivalent to a fresh, 48-week, parallel-group trial. Carryover effects (residual drug in the system) and expectation effects (patients and doctors knowing who switched to what) are real limitations on how much weight this evidence can carry on its own.

Source: Phase 2 published results.

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11 What Changes Going Into Phase 3

This slide is a bridge: it lays out, point by point, exactly how the systemic Phase 3 trial (NCT06104124) differs from the Phase 2 trial the positive evidence above is based on — and what each change implies for the forecast.

Factor	Phase 2 (systemic)	Phase 3 (systemic)	What it implies
Total randomized	74	651	More statistical precision — but not proportionally larger effect. A bigger trial reduces noise, it doesn't make the drug work better.
Active vs. control (one comparison)	36 / 38	~217 / 217 (assumed)	Roughly a 6× increase in each arm's size — the registry does not confirm the allocation is exactly equal, so this is an assumption.
Primary assessment timepoint	Day 169	Week 48 (~day 336)	A much longer follow-up window, which both allows more durability testing and gives the placebo group more time to catch up (a bigger “placebo-response window”).
Active dose arms	One dose	Two dose arms	Dose selection and how the statistical testing budget is split between the two doses (multiplicity) now matter — discussed next on Slide 12.
ESSDAI entry floor	≥ 5	≥ 5	Unchanged — there is no evidence the eligibility bar was loosened.
Baseline patient mix / disease-domain profile	Directly observed	Not established	A genuine open question: does the larger Phase 3 population resemble the Phase 2 population closely enough for the effect to transfer? This is not confirmed either way.

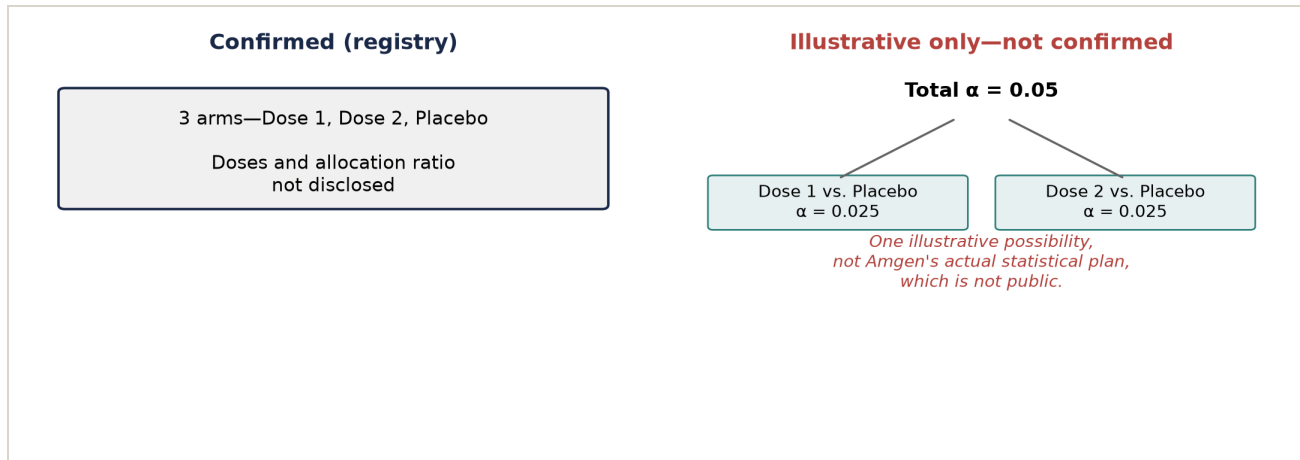
Rows shown in red in the original slide indicate items that are **assumed or not established** by the public registry, as opposed to confirmed facts. The report is explicit about which numbers are known versus estimated throughout.

Source: project analysis notes and Phase 2 baseline data. Allocation ratio is assumed equal between arms; this is not confirmed by the trial registry.

PART I · SLIDE 12 OF 26

12 Dose Selection: PK/PD Modeling

This slide explains the scientific basis for how the Phase 3 doses were likely chosen, and separately, flags an open statistical question about how the two-dose design will be analyzed.



Note on the chart above: in the source deck this diagram is filed under an asset originally labeled for a different slide number, but its content — a population pharmacokinetic/pharmacodynamic (PK/PD) model and an illustrative multiple-testing scheme — matches this slide's subject matter, so it is presented here.

A population PK/PD model pooled data across four earlier trials (healthy volunteers, rheumatoid arthritis patients, and Sjögren's patients), relating how much drug is in a patient's system to how much it suppresses the relevant immune biomarkers. One real, population-specific finding from that model: **Sjögren's patients clear the drug more slowly** than healthy volunteers or RA patients do — meaning the same dose stays active longer in this patient population. The model supports the idea that the two doses chosen for Phase 3 should sustain drug exposure above the biomarker threshold associated with immune suppression. It is important to be precise about what that proves: it is a **pharmacodynamic** target (the drug engages its biological target), not a guarantee of clinical efficacy.

The multiplicity question: because Phase 3 tests two active doses against one shared placebo group, the statistical significance threshold has to be split or adjusted somehow to avoid inflating the false-positive rate from running two comparisons. The slide shows one *illustrative* possibility — splitting a total $\alpha=0.05$ budget into $\alpha=0.025$ for each dose-vs-placebo comparison — but is explicit that this is **not Amgen's actual, disclosed statistical plan**, which is not public. This is exactly the kind of testing-procedure detail that could turn an otherwise-positive-looking result into a technical miss if the real threshold turns out to be stricter than assumed.

Source: project analysis notes on PK/PD modeling and an illustrative scenario model (not Amgen's actual statistical analysis plan).

13 Competitor Precedent: Ianalumab

Before trusting that a systemic disease-activity endpoint can even be won in a modern Phase 3 Sjögren’s trial, it helps to check whether anyone has actually done it. This slide looks at ianalumab, a competitor drug (different mechanism, described on Slide 5) that has already reported individual-arm Phase 3 results.

Trial / regimen	Effect vs. placebo (ESSDAI points)	95% CI	p-value
NEPTUNUS-1, monthly dosing	−1.3	−2.6 to 0.0	0.0496
NEPTUNUS-2, monthly dosing	−1.0	−2.0 to 0.0	0.041
NEPTUNUS-2, every-3-months dosing	not significant (CI not reported)		—

The key takeaway: monthly dosing hit statistical significance in both trials — barely, in both cases (p just under 0.05) — while the less-frequent, every-3-months regimen did not. This is genuinely useful precedent: it shows a week-48 ESSDAI benefit *can* be detected in a real Phase 3 trial, supporting one of the three pillars of the Slide 2 forecast. But note the effect sizes here (about 1.0–1.3 points) are noticeably **smaller** than dazodalibep’s own Phase 2 effect (2.2 points) — and these confidence intervals brush right up against zero, which is why this is framed as “validates the endpoint is feasible,” not “guarantees dazodalibep will win by a wide margin.”

Two additional notes for completeness: ianalumab’s earlier Phase 2b trial met its prespecified dose-response objective overall (4 of its 5 statistical models were significant), even though one individual pairwise comparison (the 300mg dose alone, p≈0.092) did not reach significance by itself — a reminder that one non-significant sub-comparison doesn’t mean a trial failed. Separately, the report flags an unresolved discrepancy between the trial registry (N=506 for NEPTUNUS-2) and a conference presentation (N=504) — both numbers are reported here rather than the report silently picking one.

Source: competitor analysis of ianalumab’s NEPTUNUS Phase 3 program and Phase 2b results.

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14 The Broader Competitive Landscape

Zooming out further, this slide surveys results from several other drugs tested in Sjögren's disease — both same-pathway drugs and drugs with unrelated mechanisms — to stress-test the assumptions behind the forecast.

Evidence	Result	What it contributes
Iscalimab, systemic cohort (N=173)	Met dose-response objective	Supports the plausibility of the CD40 pathway (same pathway as dazodalibep, opposite side)
Iscalimab, symptom cohort (N=100)	ESSPRI -0.57; p=0.12	Symptom benefit is <i>not</i> automatically assured just because the pathway works
Abatacept, ASAP-III trial	Missed continuous ESSDAI primary	Plausible biology can still fail clinically — a caution against overconfidence
Rituximab, TRACTISS trial	Missed symptom-VAS responder primary	How you define the endpoint and who you enroll matters a great deal
Tocilizumab, ETAP trial	Missed composite-response primary	A different kind of endpoint construct (composite, not continuous ESSDAI) — not directly comparable

The point of this slide is not to compute a batting average for the drug class as a whole — the report explicitly warns against that, because these trials differ too much in design, endpoint, and population to average together meaningfully. Instead, it is used to sanity-check specific assumptions: iscalimab shows the same CD40 pathway can move a systemic endpoint (supporting the mechanism story from Slide 5) while simultaneously showing that pathway plausibility alone does not guarantee a symptom-score win (a caution echoed in the DASPRI risk on Slide 18). The three “missed” trials with unrelated mechanisms are there mainly as a reminder that autoimmune Sjögren's trials fail often enough, for varied reasons, that a positive call should never be treated as a formality.

Source: historical trial evidence compiled from public sources for each named drug and trial.

15 Setting the Phase 3 Effect-Size Assumption

This is the first of three “stress test” slides. Having established the Phase 2 effect was 2.2 ESSDAI points, this slide asks: what should we *assume* the true Phase 3 effect will be, given everything discussed above (bigger trial, longer follow-up, ianalumab precedent showing smaller effects, transportability uncertainty)?

Scenario	Assumed benefit (ESSDAI points)	% of Phase 2's 2.2-point effect retained	Rationale
Bull case	2.0	91%	Most of the Phase 2 signal persists into Phase 3
Base case	1.4	64%	Meaningful attenuation, without assuming the drug loses its activity
Bear case	0.6	27%	Substantial attenuation, via any combination of reduced efficacy, measurement differences, or a different patient population

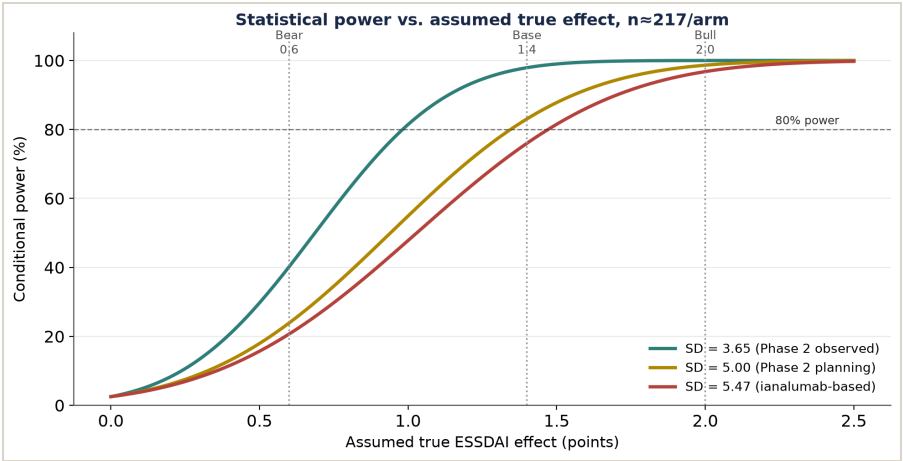
Phase 2’s own observed effect of 2.2 points is shown as a separate reference marker on the chart in the source slide — not a fourth scenario, just the anchor point everything else is measured against.

An important honesty note built into this slide: the base case of 1.4 points is explicitly labeled a **judgment call**, not a statistically fitted or empirically estimated number. That is why the analysis tests a *range* of assumptions (0.6 to 2.0) rather than betting everything on one decimal point. The report also notes that no numerical probability weights are assigned to bull/base/bear (e.g., it does not claim “60% likely to be base case”) — these are scenarios to stress-test against, not a probability-weighted forecast.

Source: project scenario model. Scenario weights are analyst judgment, not empirically estimated probabilities.

16 Statistical Power Under the Base Case

Given an assumed true effect size, how likely is the trial to actually detect it as statistically significant? This slide runs that calculation (“statistical power”) for the systemic Phase 3 trial’s single-dose comparison (n≈217 per arm), using a variance estimate borrowed from ianalumab’s Phase 3 data (effective SD=5.47, since dazodalibep’s own actual Phase 3 variance is not yet public).



Assumed true benefit (points)	Power @ $\alpha=0.05$	Power @ illustrative $\alpha=0.025$
0.6 (bear case)	21%	14%
1.0	48%	37%
1.4 (base case)	76%	66%
1.6	86%	79%
2.0 (bull case)	97%	94%

What “power” means for a general reader: it is the probability the trial will correctly detect a real effect of the assumed size, given the sample size and statistical noise involved. A power of 76% means that even if the base-case assumption (a true 1.4-point benefit) is exactly correct, there is still roughly a **1-in-4 chance the trial would come back statistically negative purely from bad luck / sampling noise** — not because the drug doesn’t work, but because trials are imperfect measuring instruments. If the stricter, multiplicity-adjusted threshold ($\alpha=0.025$) turns out to be the real one, that risk rises to roughly 1-in-3.

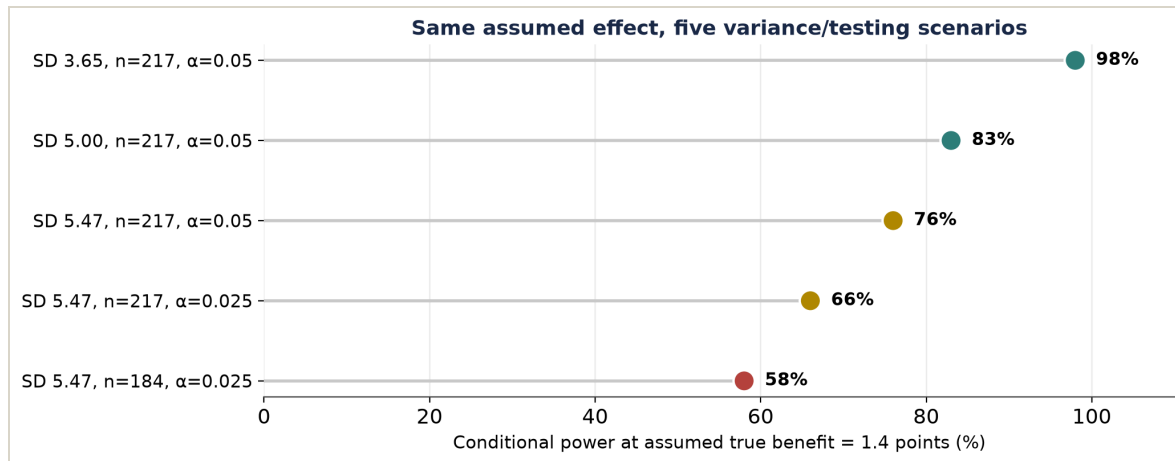
On-slide takeaway (verbatim intent): statistical success remains genuinely uncertain even if the underlying biological effect assumption is correct. This is the mathematical backbone of the “meaningful residual risk” language used in the Slide 2 executive conclusion.

Source: project scenario model. The 5.47 standard deviation is a comparator-based (ianalumab) sensitivity assumption, not a measured dazodalibep Phase 3 statistic.

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17 How Sensitive Is the Forecast to Noise?

This slide holds the assumed true benefit fixed at the base case (1.4 points) and instead varies the *other* unknowns — how noisy the data turns out to be, how strict the statistical bar is, and how much effective sample size survives dropouts and missing data — to show just how much the outcome probability can swing for the exact same underlying drug effect.



Assumption	n / arm	α	Approx. power
SD 3.65 (Phase 2's own observed variance)	217	0.05	98%
SD 5.00 (Phase 2's planning assumption)	217	0.05	83%
SD 5.47 (ianalumab-based)	217	0.05	76%
SD 5.47, stricter multiplicity-adjusted α	217	0.025	66%
SD 5.47, reduced effective sample size, stricter α	184	0.025	58%

This is the single clearest illustration in the whole report of why the forecast carries real risk. For the exact same assumed 1.4-point true benefit, the probability of a statistically positive trial ranges from a comfortable **98%** (if the data is as clean as Phase 2's own variance) all the way down to a coin-flip-adjacent **58%** (if the data is noisier, the statistical bar is stricter, and the effective sample size shrinks from missing data or dropouts). None of these are exotic or implausible assumptions — they are all grounded in real precedent (Phase 2's own numbers, the trial's own original planning assumption, and a competitor's actual Phase 3 variance).

The slide also flags that how Phase 3 handles missing data and any rescue medication (backup treatment given to patients who aren't responding) is currently unknown and directly affects which end of this range is more likely — a real, disclosed gap in what's publicly knowable right now.

Source: project scenario model.

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18 The DASPRI Risk

This slide isolates what the report considers the single biggest identified risk to the *symptomatic* trial specifically — separate from, and arguably larger than, the statistical noise risk discussed above for the systemic trial.

Question	Evidence available	Interpretation
Does dazodalibep improve reported symptoms?	Positive Phase 2 ESSPRI total and domain results (Slide 9)	Supports a positive ESSPRI lean
Does the benefit extend all the way to week 48?	Only supportive crossover data (Slide 10), no true parallel-group replication at that timepoint	Durability at week 48 remains genuinely untested
What evidence supports DASPRI specifically?	None. A full-text search of the Phase 2 paper confirms zero occurrences of DASPRI anywhere in it.	There is no dazodalibep-specific precedent for this instrument at all
What defines overall trial success?	Two co-primary outcomes (ESSPRI and DASPRI); the pass/fail rule is not publicly established	Do not assume the trial only needs one of the two to pass — it may require both

What DASPRI is: it is a newer patient-reported symptom diary, developed specifically to align with FDA guidance on patient-reported outcome instruments. It appears to have been added to the symptomatic trial for regulatory-alignment reasons — not because ESSPRI (the older, Phase-2-tested instrument) was seen as underperforming. But regardless of the reason it was added, the practical fact remains: dazodalibep has **never been tested on this instrument before**, so there is no data at all to forecast how it will perform.

This is explicitly framed as a fact about the evidence base, not a research gap that will resolve itself with more digging. No amount of re-reading the Phase 2 paper will produce DASPRI evidence that doesn't exist. Combined with the undisclosed both-pass-or-either-pass rule, this is the main reason the symptomatic trial gets a softer, more qualified “lean positive” call rather than the fuller conviction given to the systemic trial.

Source: project risk-note analysis of the symptomatic trial (NCT06245408).

19 Safety: What Phase 2 Showed

A trial can be statistically positive and still raise real safety concerns. This slide reviews the notable serious safety observations from the Phase 2 trial — explicitly framed as a selection of the most serious events, not a complete accounting of every adverse event recorded.

Observation	Population	Attribution & timing
Deep vein thrombosis (blood clot) and drug-induced liver injury, in the same participant	Population 1 (systemic)	Both investigator-attributed as related to the drug; onset 180 days after the last dose
Invasive ductal breast carcinoma	Population 2 (symptomatic)	Investigator-attributed as related; onset 139 days after the last dose
One death, cause unknown, following a COVID-19 diagnosis	Population 1 (systemic)	Reported during an earlier stage of the trial; investigators considered it unrelated to the drug. It is tabulated in the paper's later, longer follow-up safety window (46 days post-last-dose) — both facts are reported together here rather than simplified into one label.

Denominators for context: the systemic dazodalibep safety population was 36 patients initially (34 in the later follow-up stage), and the symptomatic safety set was 48 patients. In total, that is **three investigator-attributed, drug-related serious events in two participants**, plus one separately reported death considered unrelated.

How to weigh this: individual serious events in a trial this small can neither prove the drug causes these problems nor rule out a real risk — the sample size is simply too small to draw a firm conclusion either way. This is squarely a “watch this closely in the larger Phase 3 safety database” item, not a disqualifying signal and not a clean bill of health. The blood clot event is particularly worth noting given the specific design choice discussed on Slide 5 — the drug’s Fc-free engineering was intended to avoid exactly the clotting mechanism seen in earlier-generation anti-CD40L antibodies, and this event is a reminder that mitigating one known mechanism does not eliminate all clotting risk from other causes.

Source: Phase 2 published results. This is explicitly labeled as a selection of serious observations, not an exhaustive safety table.

20

The Bottom Line: Forecast Summary

This closing slide of the main deck restates the forecast plainly and, importantly, lays out how the analyst would interpret several different possible outcomes — a framework for judging the actual readout when it arrives, not just a single point prediction.

Forecast: expect a positive ESSDAI efficacy result in NCT06104124 (systemic). For NCT06245408 (symptomatic), the ESSPRI evidence is favorable, but full-trial interpretation remains genuinely unresolved because of the DASPRI and success-rule uncertainty covered on Slide 18.

If the readout shows...	...then the interpretation is
Pre-specified primary procedure met, with consistent results across both doses	Confirms the forecast; the next questions become the size of the effect and the safety profile, assessed separately
Primary met, but with a modest effect size or mixed supportive results	A statistical win, but one that calls for a more qualified clinical interpretation
Pre-specified primary procedure <i>not</i> met	The forecast was wrong — the report commits in advance to evaluating why, rather than redefining what “success” meant after the fact
Efficacy is positive, but a concerning safety pattern emerges	The efficacy call can be correct while the overall benefit-risk balance is still unfavorable — these are separate judgments

Timeline context: per the trial registry, the systemic trial’s primary completion (July 23, 2026) and study completion (August 17, 2026) are both already-actual, historical dates as of this report’s analysis cutoff. The symptomatic trial’s primary completion (estimated October 22, 2026) and study completion (estimated December 17, 2026) are still projected, not yet actual. The report is careful to note that a trial reaching “completion” in the registry is not the same as the sponsor publicly disclosing results — no specific announcement date for either readout is established or assumed here.

Source: *ClinicalTrials.gov* registry data and project analysis.

PART II

Appendix Slide Walkthrough

The presentation's six backup slides: registry facts, verified baseline data, supportive/exploratory outcomes, the statistical methodology behind the power calculations, a pointer to expanded backup material, and the full source library.

A1

Registry Facts and Unresolved Design Fields

A consolidated reference table pulling every hard fact available from the public ClinicalTrials.gov registry for both trials, side by side.

Field	Systemic (NCT06104124)	Symptomatic (NCT06245408)
Enrollment	651	434
Status	Completed	Active, not recruiting
Primary outcome(s)	ESSDAI	ESSPRI; DASPRI
Primary assessment	Week 48	Week 48
Primary completion	2026-07-23 (actual)	2026-10-22 (estimated)
Study completion	2026-08-17 (actual)	2026-12-17 (estimated)

Explicitly flagged as missing from the public registry export (i.e., fields the registry simply does not contain, so this report cannot speak to them): arm doses and dosing frequency, the allocation ratio between arms, the full eligibility criteria text, rules on keeping background therapy stable, rescue-medication rules, complete outcome-measure definitions, the protocol and statistical analysis plan (SAP), the analysis population definition, and which date fields are actual versus estimated beyond what is shown above. This table should not be read as a complete protocol summary — it is only what the registry discloses.

Source: ClinicalTrials.gov registry exports for both trials.

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A2 Baseline Patient Characteristics

This appendix reproduces the Phase 2 trial's baseline characteristics table (patient demographics and disease severity at the start of the trial, before any treatment), verified directly against the published paper's own Table 1.

Characteristic	Pop 1 Placebo (N=38)	Pop 1 DAZ (N=36)	Pop 2 Placebo (N=55)	Pop 2 DAZ (N=54)
Baseline ESSDAI, mean (SD)	10.1 (4.1)	11.4 (4.5)	2.5 (1.6)	3.1 (1.8)
Baseline ESSPRI total, mean (SD)	6.6 (1.8)	6.6 (1.6)	7.1 (1.1)	7.5 (1.5)
Dryness	6.9 (2.3)	7.3 (1.7)	6.8 (1.2)	7.1 (1.6)
Fatigue	6.8 (2.2)	7.0 (1.7)	6.8 (1.8)	7.1 (1.8)
Pain	6.1 (2.3)	5.5 (3.0)	6.3 (1.9)	6.6 (2.6)
OSDI (dry eye index)	40.0 (20.9)	48.7 (25.0)	48.6 (19.2)	48.2 (23.8)
Stimulated salivary flow, mL/min	0.97 (0.85)	0.88 (0.64)	0.83 (0.83)	1.10 (1.18)
Concomitant RA/SLE	9 (23.7%)	7 (19.4%)	7 (12.7%)	4 (7.4%)

In plain terms: the two treatment groups within each population started out reasonably similar to each other on most measures, which is what randomization is supposed to achieve. One background-therapy imbalance is worth flagging: in Population 2, use of cholinergic agonists (a class of drugs used to stimulate saliva/tear production) differed noticeably by arm (11.1% on dazodalibep vs. 27.3% on placebo) — a baseline difference that could, in principle, affect symptom-score comparisons. Population 1 also had substantial background immunosuppressive therapy use (59.5% antimalarial, 41.9% glucocorticoid, 31.1% conventional DMARD use overall).

Provenance note: these values were re-verified directly against the source paper's Table 1 image three separate times over the course of this project, after two external reviews proposed different Population 2 values for ESSPRI-total, dryness, OSDI, and salivary flow. Each re-check matched the values shown above, not the alternates proposed in those reviews (which appear to reflect a row-transposition error repeated across both). The values above are the ones to use.

Source: Phase 2 baseline characteristics data, verified against the published paper's Table 1.

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A3 Supportive and Exploratory Outcomes

Beyond the primary endpoints (ESSDAI for the systemic population, ESSPRI for the symptomatic population), the Phase 2 trial also collected several secondary and exploratory outcomes at day 169. These are presented here as supportive context, not as independently confirmatory results.

Outcome	Population	DAZ / Placebo change	p-value
ESSPRI	Systemic	−1.8 / −1.1	0.1110
FACIT-Fatigue	Systemic	+8.1 / +5.8	0.3028
FACIT-Fatigue	Symptomatic	+8.1 / +2.8	0.0095
OSDI (dry eye)	Systemic	−16.00 / −14.02	0.6583
OSDI (dry eye)	Symptomatic	−13.95 / −8.52	0.1936
Stimulated salivary flow	Systemic	+0.39 / +0.14	0.1330
Stimulated salivary flow	Symptomatic	+0.20 / −0.01	0.2170

How to read the directions: a positive FACIT-Fatigue change means improvement (less fatigue); a negative OSDI or ESSPRI change means improvement (less dryness / fewer symptoms). Of the seven rows, only the symptomatic population's FACIT-Fatigue result reaches conventional statistical significance on its own — the rest are directionally favorable to dazodalibep but did not reach significance individually. These are explicitly labeled unadjusted, exploratory p-values, meaning no correction was applied for testing multiple outcomes at once, so some caution is warranted in over-interpreting any single result here.

Source: Phase 2 published results.

A4

Statistical Methodology (Power Calculations)

For readers who want the underlying math behind the power and sensitivity analyses on Slides 16 and 17, this appendix documents the exact formulas and inputs used.

Formulas used (δ = assumed true treatment benefit; n = per-arm sample size; σ = assumed effective standard deviation):

SE = $\sigma \times \sqrt{2 / n}$

critical_z = `inverse_normal_cdf(1 -  $\alpha/2$ )`

power(δ) = `normal_cdf( $\delta/SE$  - critical_z)`

effect_for_80%_power $\approx$ (`critical_z + inverse_normal_cdf(0.80)`) $\times$ SE

σ	n	α	SE	Observed-effect significance threshold	Effect needed for ~80% power
3.65	217	0.05	0.350	0.687	0.982
5.00	217	0.05	0.480	0.941	1.345
5.47	217	0.05	0.525	1.029	1.471
5.47	217	0.025	0.525	1.177	1.619
5.47	184	0.025	0.570	1.278	1.758

Where the three standard deviation (σ) inputs come from: 3.65 is back-calculated approximately from dazodalibep’s own Phase 2 standard errors; 5.00 is the Phase 2 protocol’s own original planning assumption; 5.47 is the effective SD implied by ianalumab’s NEPTUNUS-1 reported standard error. **None of these three is the actual, undisclosed dazodalibep Phase 3 variance** — they are the best available proxies, used precisely because the real number isn’t public.

One additional illustrative calculation: scaling ianalumab NEPTUNUS-1’s standard error only for the larger dazodalibep sample size ($0.66 \times \sqrt{(137/217)} = 0.524$) implies that an *observed* 1.3-point difference at dazodalibep’s assumed sample size would carry roughly $p \approx 0.013$ — this is a precision-scaling illustration only, not an independent forecast of dazodalibep’s actual result.

Source: project scenario model documentation.

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A5 Pointer to Expanded Backup Material

The original presentation includes a “pointer” slide noting that certain underlying material was too extensive to reproduce legibly within the deck itself, and directing readers to the full source files instead. It is reproduced here for completeness of this report’s index of everything referenced in the original presentation:

- **Ianalumab — full arm-by-arm data:** the complete Phase 2b dose-response model results plus every individual pairwise dose arm, and both full NEPTUNUS Phase 3 trials including the every-3-months regimen that missed significance.
- **Historical evidence — complete table:** full detail on TRACTISS, ETAP, and ASAP-III, including corrected endpoint classifications beyond the summary shown on Slide 14.
- **Phase 2 safety — full adverse event table:** every adverse event recorded in the Phase 2 trial, not only the headline serious events discussed on Slide 19.

Source: this is a pointer to underlying project source files, not new data in itself.

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A6 Complete Source Library

The full list of primary sources relied on across the presentation, reproduced here as a citation appendix.

1. St. Clair EW et al. “CD40 ligand antagonist dazodalibep in Sjögren’s disease: a randomized, double-blinded, placebo-controlled, phase 2 trial.” *Nature Medicine* 30, 1583–1592 (2024). DOI: 10.1038/s41591-024-03009-3. Open access (CC-BY 4.0).
2. Novartis Immunology Pipeline Event, October 2025 — NEPTUNUS individual-arm efficacy data (ianalumab).
3. ClinicalTrials.gov, NCT06104124 — systemic Phase 3 trial registry entry.
4. Seror R et al. ESSDAI/ESSPRI minimal clinically important difference (MCID) and disease-activity states. *Annals of the Rheumatic Diseases*.
5. ClinicalTrials.gov, NCT06245408 — symptomatic Phase 3 trial registry entry.
6. Der K et al. Population PK/PD modeling of dazodalibep. ACR Convergence 2023 abstract.
7. Bowman SJ et al. “Safety and efficacy of subcutaneous ianalumab in primary Sjögren’s syndrome: phase 2b dose-finding trial.” *Lancet* 399, 161–171 (2022; online 2021). PMID 34861168.
8. Felten R et al. (ETAP trial). Interleukin-6 receptor inhibition (tocilizumab) in primary Sjögren syndrome. PMID 33208345. **THE P=0.14 FIGURE ATTRIBUTED TO THIS TRIAL IS FLAGGED IN THE UNDERLYING PROJECT NOTES AS UNCONFIRMED AGAINST THIS PRIMARY SOURCE AND SHOULD BE INDEPENDENTLY VERIFIED BEFORE FURTHER USE.**
9. Fisher BA et al. “Safety and efficacy of subcutaneous iscalimab in two distinct populations of patients with Sjögren’s disease (TWINSS).” *Lancet* (2024). PMID 39096929.
10. van Nimwegen JF et al. (ASAP-III trial). Primary randomized trial of abatacept. *Lancet Rheumatology*.
11. TRACTISS trial primary report (rituximab).

Source: full citation list compiled across the project's reference, research, and analysis materials.

BOTTOM LINE

Is the Forecast Positive or Negative?

This section answers directly the question this report was created to address: based on the evidence walked through above, is the forecast for Amgen's dazodalibep Phase 3 program **positive or negative**, and why?

OVERALL CALL

Net **POSITIVE** lean for the primary systemic trial (NCT06104124), with meaningful and quantified residual risk — not a high-conviction, low-risk call.

Net **modestly positive but unresolved** lean for the supporting symptomatic trial (NCT06245408), capped by a specific, identified evidentiary gap rather than general uncertainty.

Why the systemic trial (NCT06104124) leans positive

1. **The core clinical evidence is genuinely strong.** In Phase 2, dazodalibep beat placebo by 2.2 ESSDAI points at day 169, a statistically significant result ($p=0.0167$) that would have cleared even the stricter conventional 0.05 threshold, not just the trial's own looser 0.10 standard.
2. **The biology backs up the clinical result.** The drug measurably suppressed two relevant immune biomarkers (CXCL13, rheumatoid factor), and patients who crossed over onto the drug kept improving while patients who crossed off it partially relapsed — a coherent, mechanistically consistent story, not just a statistical fluke.
3. **The endpoint itself is provably winnable in Phase 3.** A competitor drug with a related but distinct mechanism (ianalumab) has already shown, in real Phase 3 trials, that a week-48 ESSDAI benefit against placebo can be detected and reach statistical significance — de-risking the concern that the endpoint simply cannot be won at this later timepoint.
4. **Even under a deliberately conservative base-case assumption** (a 1.4-point true benefit, only 64% of the Phase 2 effect), the trial retains 66–76% statistical power to detect it — better than a coin flip, and substantially better under less conservative but still plausible variance assumptions (as high as 98%).

Why the risk is real, not just a disclaimer

1. **The effect could shrink in a bigger, longer trial.** Phase 3 runs to week 48 instead of day 169, giving the placebo group more time to catch up, and enrolls a nearly 9× larger population whose baseline mix is not confirmed to match Phase 2's. Ianalumab's own Phase 3 effect sizes (about 1.0–1.3 points) came in meaningfully smaller than its own earlier signals, and smaller than dazodalibep's 2.2-point Phase 2 result — a real precedent for attenuation, not a hypothetical one.
2. **The statistical testing rules are not public.** With two active doses now being tested against one shared placebo, however Amgen actually splits its significance budget between them could tighten the effective bar for either dose to succeed. The 66% (versus 76%) power swing between $\alpha=0.025$ and $\alpha=0.05$ in this report's own modeling shows this is not a minor technicality.

3. **The forecast is highly sensitive to data noise that won't be known until the readout.** Holding the assumed true effect fixed at 1.4 points, this report's own sensitivity analysis shows the probability of a statistically positive result ranging all the way from 98% down to 58% depending on how noisy the actual data turns out to be and how much effective sample size survives dropouts and missing data.
4. **Responder-level evidence in Phase 2 was directionally favorable but not statistically confirmatory** at any pre-specified threshold — only a post hoc cut reached significance, meaning the average-level result should not be over-read as a guaranteed, uniformly-distributed benefit across patients.
5. **A small but real safety signal exists.** Three investigator-attributed, drug-related serious events occurred in two participants during Phase 2 (a blood clot plus liver injury in one patient, and a breast cancer diagnosis in another), alongside one separately reported unrelated death. The sample size is too small to draw firm conclusions, but this is a genuine open item for the larger Phase 3 safety database to resolve — and a reminder that a statistically positive efficacy result and a favorable overall benefit-risk balance are two different questions.

Why the symptomatic trial (NCT06245408) gets a softer call

The symptom evidence itself is, if anything, *stronger* in relative terms than the systemic evidence — the Phase 2 ESSPRI result ($p=0.0002$) is the strongest p-value anywhere in the program, with consistent, favorable results across all three symptom domains (dryness, fatigue, pain). What holds this trial back from a fuller endorsement is not weak evidence but a **specific, identified evidentiary gap**: the trial's second co-primary endpoint, DASPRI, has zero precedent in any dazodalibep trial to date, and the rule for what combination of ESSPRI and DASPRI results counts as overall trial success has not been publicly disclosed. This is a real fact about the evidence base, not a gap that further analysis of existing data could close. It is why this trial is forecast as “lean positive” rather than carrying the fuller conviction given to the systemic trial.

Bottom line

Weighing the evidence and the risks together, the net call is that **dazodalibep is more likely than not to deliver a statistically positive systemic ESSDAI result** in NCT06104124, on the strength of a genuinely strong, biologically coherent Phase 2 signal and external precedent that the endpoint is achievable in Phase 3 — but this is a probabilistic lean under real, quantified uncertainty (roughly two-thirds to three-quarters power under this report's own base-case and sensitivity assumptions), not a near-certain outcome. A negative primary result would not be a shocking surprise given how sensitive the power calculations are to unknowns like true effect size, data variance, and the actual statistical testing plan. The symptomatic trial is directionally favorable on the evidence that exists, but carries a distinct, disclosed risk (DASPRI) that has no historical data to resolve one way or the other before the readout itself arrives. Both calls should be read as informed, evidence-based forecasts under genuine uncertainty — exactly the framing the original presentation adopts throughout — rather than as confident predictions of certain success.

REPORT APPENDIX A

Full Glossary

ESSDAI

EULAR Sjögren's Syndrome Disease Activity Index. A clinician-scored measure of systemic (organ-level) disease activity across weighted body-system domains. Higher = worse; a larger negative change = more improvement.

ESSPRI

EULAR Sjögren's Syndrome Patient Reported Index. A patient-reported questionnaire covering dryness, fatigue, and pain. Higher = worse; a larger negative change = more improvement.

DASPRI

A newer patient-reported symptom diary developed to align with FDA patient-reported-outcome guidance. Has no track record in any dazodalibep trial to date; used as a co-primary endpoint in the symptomatic Phase 3 trial (NCT06245408).

CD40L / CD40

A pair of proteins on immune cells (CD40L on T-cells, CD40 on B-cells and other cells) whose interaction (“costimulation”) helps drive immune activation. Dazodalibep blocks CD40L; iscalimab blocks CD40; both interrupt the same signal from opposite sides.

Fc-free / Tn3 scaffold

An antibody-engineering design choice used for dazodalibep that avoids the Fc portion of a normal antibody. This is intended to avoid a blood-clotting risk (platelet cross-linking) seen with earlier, first-generation anti-CD40L antibodies.

Adjusted mean / adjusted difference

A treatment-effect estimate that has been statistically adjusted (e.g., for baseline differences between groups) rather than a simple raw average difference.

Confidence interval (CI)

A range of values that is likely to contain the true treatment effect. A 95% CI is wider (more conservative) than a 90% CI for the same data. If a CI for a difference does not cross zero, the result is statistically significant at the corresponding threshold.

p-value

A statistical measure of how likely a result this extreme (or more extreme) would be if there were truly no treatment effect. Conventionally, $p < 0.05$ is treated as “statistically significant,” though this trial program in places uses a 0.10 threshold (see α below).

 α (alpha) / significance threshold

The pre-specified false-positive rate a trial is designed to control. $\alpha=0.05$ is the conventional standard (95% confidence); $\alpha=0.10$ (90% confidence) is a looser, more permissive threshold sometimes used in smaller early-phase trials.

Statistical power

The probability a trial will correctly detect a real treatment effect of an assumed size, given its sample size and expected data variability. Lower power means a higher chance of a false-negative (missing a real effect) result.

Multiplicity / multiplicity adjustment

The statistical problem of controlling the overall false-positive rate when a trial runs more than one comparison (e.g., two active doses vs. one placebo). Typically handled by splitting the overall significance budget across comparisons, which makes each individual comparison's bar stricter.

Co-primary endpoint

One of two (or more) outcome measures that a trial is designed to test as equally central to defining trial success, as opposed to a single primary endpoint with secondary/supportive measures.

Post hoc

An analysis decided on and performed after seeing the data, rather than being specified in the trial's design in advance. Post hoc findings are considered suggestive/hypothesis-generating rather than confirmatory.

Responder analysis

An analysis of how many individual patients crossed a pre-defined improvement threshold, as opposed to looking only at the group average change.

Crossover

A trial design element where, after an initial randomized comparison period, patients switch treatment groups (e.g., placebo patients move onto active drug). Useful for durability signals but not equivalent to a fresh randomized comparison.

Biomarker (CXCL13, rheumatoid factor)

Substances measurable in blood that reflect underlying disease/immune biology. Used here as lab-based evidence that a drug is biologically active, independent of clinical or patient-reported outcomes.

PK/PD modeling

Pharmacokinetic/pharmacodynamic modeling — mathematical modeling of how a drug moves through the body (PK) and how much biological effect a given exposure produces (PD). Used to support dose selection.

Primary completion date / study completion date

Registry terms: primary completion is when the last participant was assessed for the primary outcome; study completion is when the entire trial (including any secondary follow-up) concluded. Neither implies public disclosure of results on that date.

REPORT APPENDIX B

Complete Source List

The full list of primary sources relied on across the presentation, reproduced here as a citation appendix.

1. St. Clair EW et al. “CD40 ligand antagonist dazodalibep in Sjögren’s disease: a randomized, double-blinded, placebo-controlled, phase 2 trial.” *Nature Medicine* 30, 1583–1592 (2024). DOI: 10.1038/s41591-024-03009-3. Open access (CC-BY 4.0).
2. Novartis Immunology Pipeline Event, October 2025 — NEPTUNUS individual-arm efficacy data (ianalumab).
3. ClinicalTrials.gov, NCT06104124 — systemic Phase 3 trial registry entry.
4. Seror R et al. ESSDAI/ESSPRI minimal clinically important difference (MCID) and disease-activity states. *Annals of the Rheumatic Diseases*.
5. ClinicalTrials.gov, NCT06245408 — symptomatic Phase 3 trial registry entry.
6. Der K et al. Population PK/PD modeling of dazodalibep. ACR Convergence 2023 abstract.
7. Bowman SJ et al. “Safety and efficacy of subcutaneous ianalumab in primary Sjögren’s syndrome: phase 2b dose-finding trial.” *Lancet* 399, 161–171 (2022; online 2021). PMID 34861168.
8. Felten R et al. (ETAP trial). Interleukin-6 receptor inhibition (tocilizumab) in primary Sjögren syndrome. PMID 33208345. **THE P=0.14 FIGURE ATTRIBUTED TO THIS TRIAL IS FLAGGED IN THE UNDERLYING PROJECT NOTES AS UNCONFIRMED AGAINST THIS PRIMARY SOURCE AND SHOULD BE INDEPENDENTLY VERIFIED BEFORE FURTHER USE.**
9. Fisher BA et al. “Safety and efficacy of subcutaneous iscalimab in two distinct populations of patients with Sjögren’s disease (TWINSS).” *Lancet* (2024). PMID 39096929.
10. van Nimwegen JF et al. (ASAP-III trial). Primary randomized trial of abatacept. *Lancet Rheumatology*.
11. TRACTISS trial primary report (rituximab).

Source: full citation list compiled across the project's reference, research, and analysis materials.

REPORT APPENDIX C

Methodology & Limitations of This Report

This section documents how this report itself was produced, and the limitations that apply to it.

What this report is

This report is a plain-language explanatory companion to a 26-slide equity-research presentation. Every metric, table value, chart, and source citation in this report is carried over directly from that presentation's content and its underlying project analysis files — no numbers have been independently recalculated, re-derived, or estimated by this report. Explanatory language, structure, the background primer, the glossary, and the synthesized conclusion are new material written for this report; the underlying facts and figures are not.

What this report is not

This report is not independent medical, investment, or statistical advice, and should not be relied upon as such. The forecast summarized here is a qualitative, directional analyst judgment about the likelihood of specific trials meeting their pre-specified statistical endpoints — it is explicitly not a numerically calibrated probability (e.g., it does not claim a specific percent chance of trial success), and it is not a clinical or regulatory assessment of the drug's overall benefit-risk profile, which would require information (the full protocol, statistical analysis plan, complete safety database, and eventual trial results) not publicly available as of the analysis cutoff date of September 15, 2026.

Known limitations carried over from the underlying analysis

- Several key Phase 3 inputs (exact allocation ratios, the actual statistical testing/multiplicity plan, the true variance of the Phase 3 data, and the DASPRI/ESSPRI success rule for the symptomatic trial) are not publicly disclosed and are explicitly treated as assumptions or open questions throughout, not as known facts.
- Comparator evidence (ianalumab, iscalimab, and other drugs) is used for context and precedent, not as a direct predictor of dazodalibep's own results — different drugs, mechanisms, and trial designs are not directly interchangeable.
- Phase 2 safety observations come from a small trial and cannot establish or rule out causation for the serious events discussed at Slide 19.
- One source discrepancy (ianalumab NEPTUNUS-2's enrollment, reported as both N=506 and N=504 across different sources) remains unresolved and is reported as such rather than arbitrarily resolved.
- One citation (the ETAP trial's $p=0.14$ figure) is flagged in the underlying project notes as unconfirmed against its primary source and should be independently verified before further reliance.